Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Course Plan

Course Name	:	Applied Linear Algebra
Course Code	:	AME 5101
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Dr. Devi Prasad M
Name of the Program Coordinator	:	Dr. Richa Sharma

Date:	10/8/2026
Signature of Program Coordinator with Date	

Date:	10/8/2026
Signature of Course Coordinator with Date	

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Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning), programare as follows

PEO No.	Education Objective
PEO 1	Acquire solid mathematical and computational skills essential for understanding, applying, and developing modern artificial intelligence and machine learning algorithms.
PEO 2	Produce industry-ready graduates with practical experience in structuring machine learning projects using state-of-the-art software.
PEO 3	Address multi-disciplinary challenges through coursework and projects by adapting to the rapidly advancing developments in artificial intelligence and machine learning approaches.

Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning), graduates will be able to:

PO 1	An ability to independently carry out research/investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	Demonstrate a degree of mastery over the area as per the specialization of the program which should be at a level higher than the requirements in an appropriate bachelor program.
PO 4	Identify appropriate and efficient algorithmic approaches for solving real-life challenges using artificial intelligence and machine learning principles and state-of-the art software prevalent in industry and academia.
PO 5	Develop teamwork and leadership skills for addressing challenges of social importance for sustainable societal development using ethical artificial intelligence and machine learning approaches.

1. Course Plan

1.1 Primary Information

Course Name	:	Applied Linear Algebra [AME 5101]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	Basic Algebra and Ca;culus
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No.ofContact Hours	Program Outcomes (POs)	BL
CO1	Apply vector operations for similarity learning.	10	PO3	3
CO2	Analyse matrix representations in AI & ML models.	14	PO4,PO3	4
CO3	Apply matrix calculus for optimization in learning systems.	12	PO3	3

1.3 Assessment Plan

Components	Mid semester exam (MSE)	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	2 assignments. 15 days each.	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Applying and Analyzing	Applying and Analyzing	Applying and Analyzing
Pattern	Answer all 5 questions of 10 marks each. Each question may have 2 parts of equal marks.	Apply and analyze solutions to real-world problems.	Answer all 10 questions of 10 marks each. Each question may have 2 parts of equal marks.
Schedule	As per academic calendar.	September 15 and October 10	As per academic calendar.
Topics covered	Vectos, matrics, GPU/TPU support for deep learning systems.	Vectors, matrices, efficient solutions on GPU/TPUs.	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.

1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Conceptual introduction to vector spaces and vector operations. Scalar product.	CO1
L2	Introduction to matrices. Types of matrices. Operations. Matric-vector multiplication.	CO2
L3	Inner and outer products. Norm of a vector. Distance - standard deviation. Standardizations versus normalization.	CO1
L4	Linear independence and basis vectors.	CO2
L5	High-dimensional vector embeddings as in Word2Vec and LLMs.	CO2
L6	Linear operations on embeddings: Finding relationships, combining and isolating concepts, and analogies.	CO1
L7	Systems of linear equations: over and under-determined systems.	CO1
L8	Gram-Schmidt algorithm.	CO3
L9	QR-factorization.	CO2
L10	Solving linear equations.	CO1
L11	Applications in deep learning - Introduction to sensitivity and gradient in one dimension.	CO3

L12	Chain rule. Matrix derivatives.	CO3
L13	Gradient descent.	CO2
L14	Training the artificical neural networks - backward propagation of errors.	CO2
L15	Backpropagation algorithm.	CO3
L16	Introduction to Attention mechanism.	CO1
L17	Impementation details of attention mechanism in large language models.	CO3
L18	Impementation details of attention mechanism in large language models.	CO3
L19	Tensor cores and GPU architectures	CO2
L20	Tensor cores and GPU architectures	CO2
L21	Least squares: problem motivation and examples.	CO2
L22	Solving least squares problems - least square data fitting.	CO1
L23	Solving least squares problems - least square data fitting.	CO3
L24	Maximizing variance and minimizing projection error.	CO1
L25	Minimizing projection error, and identifying principal components visually.	CO3

L26	Mean centering and covariance matrices.	CO2
L27	Covariance matrices, eigenvectors, and eigenvalues.	CO3
L28	Mean centering, covariance matrices, eigenvectors, and eigenvalues - Data Preparation and Relations.	CO3
L29	Mean centering, covariance matrices, eigenvectors, and eigenvalues - The Transformation Engine.	CO3
L30	Feature scaling (standardization), eigendecomposition vs. Singular Value Decomposition (SVD)	CO3
L31	Applying PCA to solve machine learning bottlenecks	CO1
L32	Applying PCA to solve machine learning bottlenecks	CO1
L33	Loss of feature interpretability, sensitivity to outliers, and the limitation of linear assumptions	CO2
L34	Batch Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-batch Gradient Descent. Vectorization efficiency, memory constraints, convergence speed, and escaping local minima.	CO2
L35	Sigmoid with Binary Cross-Entropy vs. Softmax with Categorical Cross-Entropy. Mapping out the backward pass differences between two-class and many-class problems.	CO2
L36	Linear/Identity activation functions at the output layer and Mean Squared Error (MSE) loss. Comparing gradient behavior and scale stability between regression and classification losses	CO2

1.5 References

1. Introduction to Applied Linear Algebra, Vectors, Matrices, and Least Squares, Stephen Boyd & Lieven Vandenberghe, Cambridge University Press, 1st Edition, 2018. Available online at http://vmls-book.stanford.edu/vmls.pdf.
2. Linear Algebra and Learning from Data, Gilbert Strang, Cambridge Uni. Press, 1st Edition, 2019.
3. IBM AI Engineering Professional Certificate (https://www.coursera.org/professional- certificates/ai-engineer).
4. Deep Learning with Python, François Chollet, Manning Publications, 2nd Edition, 2021.
5. Neural Networks and Deep Learning, Michael A. Nielsen, Determination Press, 2015. Available online at http://neuralnetworksanddeeplearning.com/
6. Matrix Methods: Applied Linear Algebra, Richard Bronson and Gabriel B. Costa, Academic Press, 3rd Edition, 2008.
7. Introduction to computational thinking (https://computationalthinking.mit.edu/Fall24/).

1.6 Other Resources (Online, Text, Multimedia, etc.)

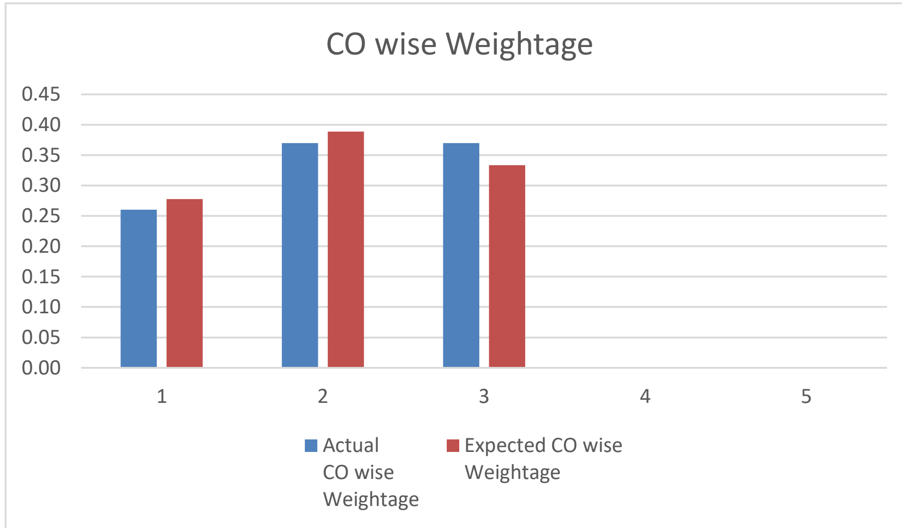
1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.

1.7 Course Timetable

I Semester						Room: LG01-LH12			
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday		ALA							
Tuesday									
Wednesday		ALA							
Thursday									
Friday		ALA							
Saturday									

1.8 Assessment Plan

COs		Marks & Weightage				
CO No.	**CO Name**	**MSE Marks (Max. 50)**	**Assignment Marks (Max. 20)**	**ESE Marks (Max. 100)**	**Actual CO wise Weightage**	**Expected CO wise Weightage**
CO1	Apply vector operations for similarity learning.	10	10	20	**0.26**	**0.28**
CO2	Analyse matrix representations in AI & ML models.	20	5	40	**0.37**	**0.39**
CO3	Apply matrix calculus for optimization in learning systems.	20	5	40	**0.37**	**0.33**
	Marks (weightage)	**0.3**	**0.2**	**0.5**	**1.00**	**1.00**



Note:

- In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes the performances in class participation, assignment work, class tests, MSE, quizzes etc.
- ESE for this course is conducted for a maximum of 100 and the same will be scaled down to 50 for grading.
- ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

Weightage for CO1 = (MSE marks for CO1 / 1.6666 + Assignment marks for CO1/1.0 + ESE marks for CO1 / 2)/100

1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

Sl. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	MSE	0.3	1	• Performance is measured using MSE attainment level. • Reference: question paper and answer scheme. • MSE is assessed for a maximum of 50 marks and scaled down to 30 marks.
2	Assignments	0.2	2	• Performance is measured using assignments/quiz attainment level. • Assignments/quiz are evaluated for a maximum of 20 marks.
3	ESE	0.5	1	• Performance is measured using ESE attainment level. • Reference: question paper and answer scheme. • ESE is assessed for a maximum of 100 marks and scaled down to 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1			Y		
CO2			Y	Y	
CO3			Y		
Average Articulation Level			*	*	